



fellows fellowssc <fellows@science-corps.org>

New Science Corps Fellow Application: Dina Traykova

1 message

Science Corps Web <eduversumdev@gmail.com>

Fri, Jul 31, 2020 at 9:41 AM

To: fellows@science-corps.org

Name

Dina Traykova

Gender

Female

Nationality

Bulgarian

Emaildina.traykova@physics.ox.ac.uk**When did or will you finish your PhD?**

April 2020

City, state, and country of residence

Oxford, UK

University education

DPhil in Astrophysics, University of Oxford, 2017-2020

MSci in Theoretical Physics (integrated bachelor's and master's degree), University College London, 2013-2017

What are your reasons for applying to be a Science Corps Fellow? (max. 200 words)

I am applying to be a Science Corps Fellow as I believe that the best way to further the development of science is to inspire and recruit young minds from a wider range of backgrounds. This begins with improving access to scientific education for children in communities that may not be sufficiently equipped to do so themselves.

I went to a school which could not afford scientific equipment, despite being the top science school in my region and such resources were even less available to others. As a result I saw how even with a solid theoretical education, many talented children would quickly lose their

motivation to study science without the opportunity to cultivate their interest through practical exploration. This experience makes me personally motivated to work on inspiring curiosity in children who come from more disadvantaged backgrounds than myself. This would be a unique opportunity for me to expand on my experience in teaching and designing educational content. I also believe my academic knowledge, combined with past experience in teaching and outreach, make me well-suited to this program and confident I can tackle the challenges involved whilst creating an enjoyable experience for myself and the children.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship (max. 200 words)

During my PhD I have lead tutorials for a third year undergraduate course in General Relativity. During these tutorials I worked with a group of students, marking and going through their homework problem sets, as well as helping them to cover gaps in their knowledge and develop their understanding of the material in order for them to prepare for the exam in that subject.

I have also volunteered in departmental outreach, where different groups of primary or high school students would visit the Astrophysics department and participate in a number of activities designed to inspire them towards studying Astrophysics. Some of the activities I have lead include giving the students short talks on various subjects within Astrophysics, guiding them through exercises to demonstrate ideas discussed and showing them how a telescope is operated using a computer to obtain scientific data.

In addition, as a member of the Bulgarian red cross youth I helped design and run events with educational and recreational activities and workshops on a variety of topics for children of different ages in schools, nurseries and a local orphanage.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia
Doing science in industry
In science outreach
Other

If other, please specify.

Data scientists in environmental or health care tech

Expand on your selection above and describe where you imagine yourself professionally in the near (2 year) and long-term (10 year) future (max. 200 words)

My shorter term plans for my career after I finish my PhD are to continue my research in a junior academic post-doc position while maintaining and broadening my involvement in scientific outreach and education.

In the longer term, as an academic researcher I would like to lead my own research group in the fields of numerical general relativity and modified gravity. I imagine myself in this role as continuing to be involved in inspiring and educating young scientists as a lecturer and as a research supervisor.

However, I can also see myself applying my computational, statistical and general research skills to other pursuits, such as a data scientist role in industry or at a tech start up. My ideal such position would be in the fields of environmental preservation or health care, where I can apply the expertise and experience I have acquired from research to tackling real-world issues, which I would find extremely fulfilling.

In either case, I would like to stay actively involved in organising and leading science outreach far into the future.

CV/Resumé

[DTraykova_CV.pdf](#)

How did you find out about Science Corps?

Some other mailing list

Sent from Science Corps